

Armaan Sandhu

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Education

Worcester Polytechnic Institute <i>Worcester, MA</i>	2025 - Present
Computer Science GPA: 4.0 Dean's List (Fall 2025, Spring 2026) Expected May 2029	
Carmel Junior College <i>Jamshedpur, India</i>	2010 - 2025
High School Diploma 97% (Unweighted) Valedictorian, Class of 2025	

Technical Skills

Languages: Python, C, C++, Rust, Zig, Java, x86-64 Assembly, JavaScript

Tools: Git, GDB, Linux, Bash, Valgrind

Concepts: Systems Programming, Concurrent/Async Programming, Binary Analysis, Query Planning, AI Infrastructure, Open Source Development

Open Source Contributions

lance/lancedb (rust)	May 2026 - June 2026
- Eliminated SQL serialization overhead in query planning by implementing native DataFusion expression support across the core library and Python SDK. - Resolved thread-pool contention under concurrent search workloads by isolating blocking operations on a dedicated executor. - Hardened the Bedrock embedding provider against production failures by replacing crash-prone code paths with structured error handling. - Improved developer experience by surfacing actionable errors for invalid schema definitions that previously produced opaque crashes.	
Lightpanda Browser Engine (Zig)	May 2026 - June 2026
- Implemented the W3C File API from specification in a production headless browser engine, enabling standards-compliant file handling for web automation. - Added full file input support including programmatic file injection via the Chrome DevTools Protocol, enabling automated file upload workflows.	
CPython	May 2026
Resolved build system defects in the CPython Cygwin port, improving cross-platform compiler portability and configuration correctness.	
LiteLLM Proxy (Python)	June 2026
- Restored compatibility with OpenAI-style clients by fixing budget enforcement logic that incorrectly blocked model discovery on exhausted budgets. - Introduced an opt-out setting for phantom budget accumulation affecting high-traffic Redis-backed deployments.	
sqlx (Rust)	June 2026
Fixed a type inference bug in the query macro that caused runtime crashes on MySQL zero-date columns, aligning compile-time behavior with the existing runtime contract.	

Experience

Open Source Author & MCP Deployment Consultant <i>mcp-persist / MCP Python SDK deployments</i>	May 2026 - Present
- Built and maintain mcp-persist, a production-grade persistence layer for MCP deployments supporting Redis, SQLite, and PostgreSQL backends: actively deployed in production MCP server environments with thousands of weekly downloads. - Designed an async event-store architecture with 300+ tests and CI coverage across Python 3.10-3.13. - Support real-world MCP deployments by guiding integration, migration, and backend selection across production environments.	
Peer Learning Assistant <i>Computer Science Department, WPI</i>	August 2026 - Present
Supported undergraduate students in CS coursework through one-on-one debugging sessions, code review, and targeted explanations of systems-level concepts. Guided peers toward independent problem-solving rather than direct solutions.	
Community Advisor <i>Office of Academic Advising, WPI</i>	August 2026 - Present
Managed a residential student community and served as a frontline resource for academic and personal challenges. Helped students navigate course planning, campus resources, and the transition to college life.	

Awards & Recognition

- Valedictorian** | Carmel Junior College (Class of 2025)
- Most Outstanding Student** | Awarded for excellence in academics, leadership, and extracurriculars.
- Interschool Java Coding Competition** | 2nd Place (2023, 2024)
- Ni Dan Black Belt** | Shotokan Karate India (IOC-recognized federation)